

Solving Monge's Problem

- We already mentioned how to solve the discrete version in general:
Hungarian algorithm (sadly, $O(n^3)$)
- But there are settings where it can be solved faster!

Monge's Problem in 1D

$$\min_{\sigma \in \Sigma_n} \sum_{i=1}^n |x_i - y_{\sigma(i)}|^p, \quad p \geq 1$$

